EPIGENETICS

Module map

- Module 10: Epigenetics
- Connected lab: lab-10_epigenetics.md
- Use this deck as the visual path through the module's evidence and retrieval work.

Module 10
Epigenetics

1. Gene regulation as control
2. DNA methylation
3. Histone and chromatin change
4. Environment and expression
5. Epigenetics versus mutation

Lab evidence
lab-10_epigenetics.md

EPIGENETICS

Learning objectives

- Explain gene regulation as control of gene activity.
- Describe DNA methylation at an introductory level.
- Connect chromatin structure to gene accessibility.
- Give a careful example of environmental influence on expression.
- Distinguish epigenetic regulation from mutation.

OBJECTIVE 1
Explain gene regulation as control of gene activity.

OBJECTIVE 2
Describe DNA methylation at an introductory level.

OBJECTIVE 3
Connect chromatin structure to gene accessibility.

OBJECTIVE 4
Give a careful example of environmental influence on expression.

OBJECTIVE 5
Distinguish epigenetic regulation from mutation.

EPIGENETICS

Topic sequence

- Gene regulation as control: Cells regulate which genes are active rather than using every gene equally.
- DNA methylation: DNA methylation can reduce transcription without changing base sequence.
- Histone and chromatin change: Histone and chromatin changes alter access to DNA.
- Environment and expression: Environmental conditions can influence gene expression patterns.
- Epigenetics versus mutation: Epigenetic change affects gene activity, while mutation changes DNA sequence.

01
Gene regulation as control
Cells regulate which genes are active rather than using every gene equally.

02
DNA methylation
DNA methylation can reduce transcription without changing base sequence.

03
Histone and chromatin change
Histone and chromatin changes alter access to DNA.

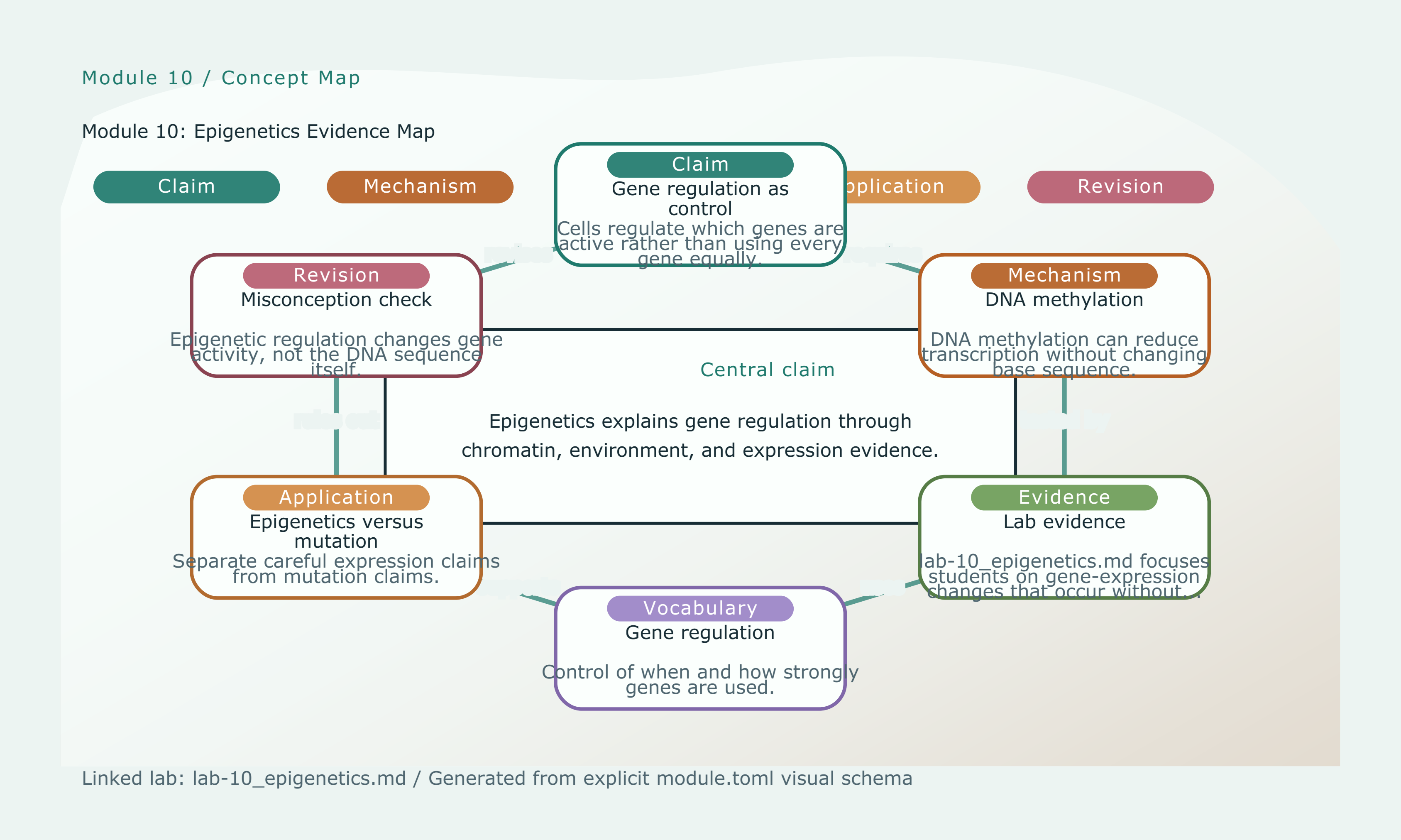
04
Environment and expression
Environmental conditions can influence gene expression patterns.

05
Epigenetics versus mutation
Epigenetic change affects gene activity, while mutation changes DNA sequence.

EPIGENETICS

Module 10: Epigenetics Evidence Map

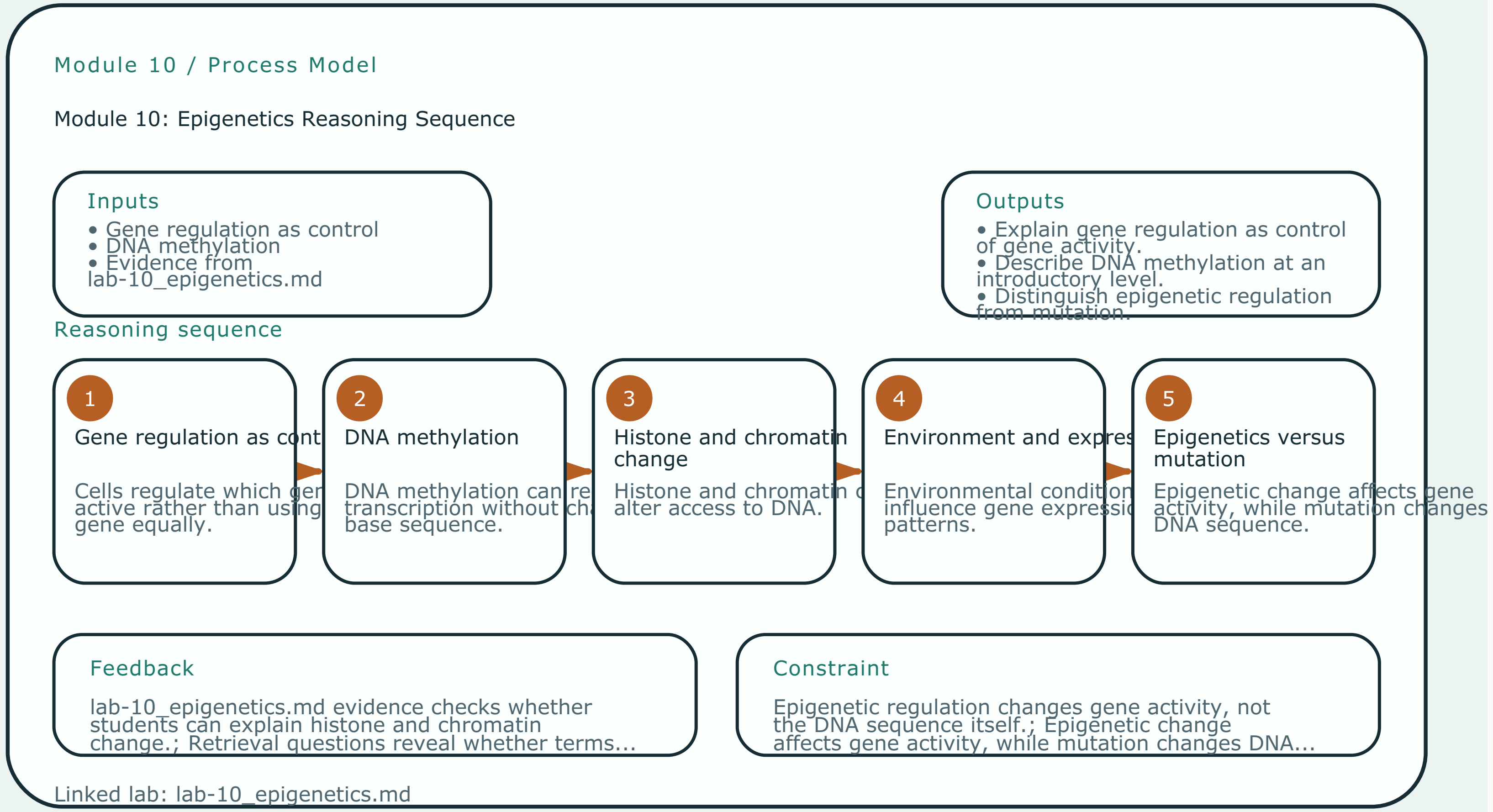
- Module focus: Epigenetics
- Central claim: Epigenetics explains gene regulation through chromatin, environment, and expression evidence.
- Key nodes to connect: Gene regulation as control, DNA methylation, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-10_epigenetics.md supplies an evidence surface for the map.



EPIGENETICS

Module 10: Epigenetics Reasoning Sequence

- Module focus: Epigenetics
- Trace the model: Gene regulation as control -> DNA methylation -> Histone and chromatin change -> Environment and expression
- Inputs become outputs: Gene regulation as control, DNA methylation -> Explain gene regulation as control of gene activity., Describe DNA methylation at an introductory level.
- Feedback check: lab-10_epigenetics.md evidence checks whether students can explain histone and chromatin change.
- Lab connection: lab-10_epigenetics.md gives students a process to observe or test.



EPIGENETICS

Terms as evidence handles

- Gene regulation: Control of when and how strongly genes are used.
- Epigenetics: Heritable or persistent expression changes without DNA sequence change.
- DNA methylation: Chemical marking of DNA often associated with reduced transcription.
- Histone: Protein around which DNA is packaged.
- Chromatin: DNA-protein complex that can be open or compact.
- Transcription factor: Protein that helps control transcription.

GENE REGULATION
Control of when and how strongly genes are used.

EPIGENETICS
Heritable or persistent expression changes without DNA sequence change.

DNA METHYLATION
Chemical marking of DNA often associated with reduced transcription.

HISTONE
Protein around which DNA is packaged.

CHROMATIN
DNA-protein complex that can be open or compact.

TRANSCRIPTION FACTOR
Protein that helps control transcription.

EPIGENETICS

Lab connection

- Lab file: lab-10_epigenetics.md
- Lab evidence should test or illustrate: Histone and chromatin change
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

QUESTION
Gene regulation as control

EVIDENCE
lab-10_epigenetics.md

CLAIM
Explain gene regulation as control of gene activity.

EPIGENETICS

Contrast check

- Do not stop at: Cells regulate which genes are active rather than using every gene equally.
- Stronger explanation: Epigenetic change affects gene activity, while mutation changes DNA sequence.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE

Cells regulate which genes are active rather than using every gene equally.

DEEPER

Epigenetic change affects gene activity, while mutation changes DNA sequence.

EPIGENETICS

Module 10: Epigenetics Retrieval and Lab Check

- Module focus: Epigenetics
- Retrieval prompt: Why do cells regulate genes?
- Answer check: Explain gene regulation as control of gene activity.
- Required terms: Gene regulation, Epigenetics, DNA methylation
- Lab connection: lab-10_epigenetics.md; lab-10_epigenetics.md supplies evidence for gene-expression changes that occur without changing DNA sequence.

Module 10 / Retrieval Card

Module 10: Epigenetics Retrieval and Lab Check

Routine
Cover notes -> answer aloud -> cite evidence -> revise

1 Cover notes | 2 Answer aloud | 3 Cite evidence | 4 Revise

Why do cells regulate genes?
Check: Explain gene regulation as control of gene activity.

How can two cells with the same DNA act differently?
Check: Describe DNA methylation at an introductory level.

What is DNA methylation?
Check: Connect chromatin structure to gene accessibility.

How can chromatin structure affect transcription?
Check: Give a careful example of environmental influence on expression.

Terms to use
Gene regulation, Epigenetics, DNA methylation, Histone, Chromatin

Lab connection
lab-10_epigenetics.md supplies evidence for gene-expression changes that occur without changing DNA...

Intro biology routine: claim, evidence, reasoning, revision.

EPIGENETICS

Practice quiz bridge

- 1. How does epigenetic regulation differ from a mutation?
- 2. DNA methylation is usually associated with:
- 3. When chromatin is packed tightly, genes in that region are generally:
- 4. Two genetically identical twins come to differ in a trait over their lifetimes. A good explanation is that:

QUESTION 1

Answer first

How does epigenetic regulation differ from a mutation?

QUESTION 2

Answer first

DNA methylation is usually associated with:

QUESTION 3

Answer first

When chromatin is packed tightly, genes in that region are generally:

QUESTION 4

Answer first

Two genetically identical twins come to differ in a trait over their lifetimes. A good explanation is that:

EPIGENETICS

Synthesis and exit ticket

- Exit claim: explain gene regulation as control using evidence from lab-10_epigenetics.md.
- Must include: Gene regulation, Epigenetics, DNA methylation.
- Revision prompt: what evidence would change your explanation?

CLAIM
Gene regulation as control

EVIDENCE
lab-10_epigenetics.md

REVISION
What would change your mind?